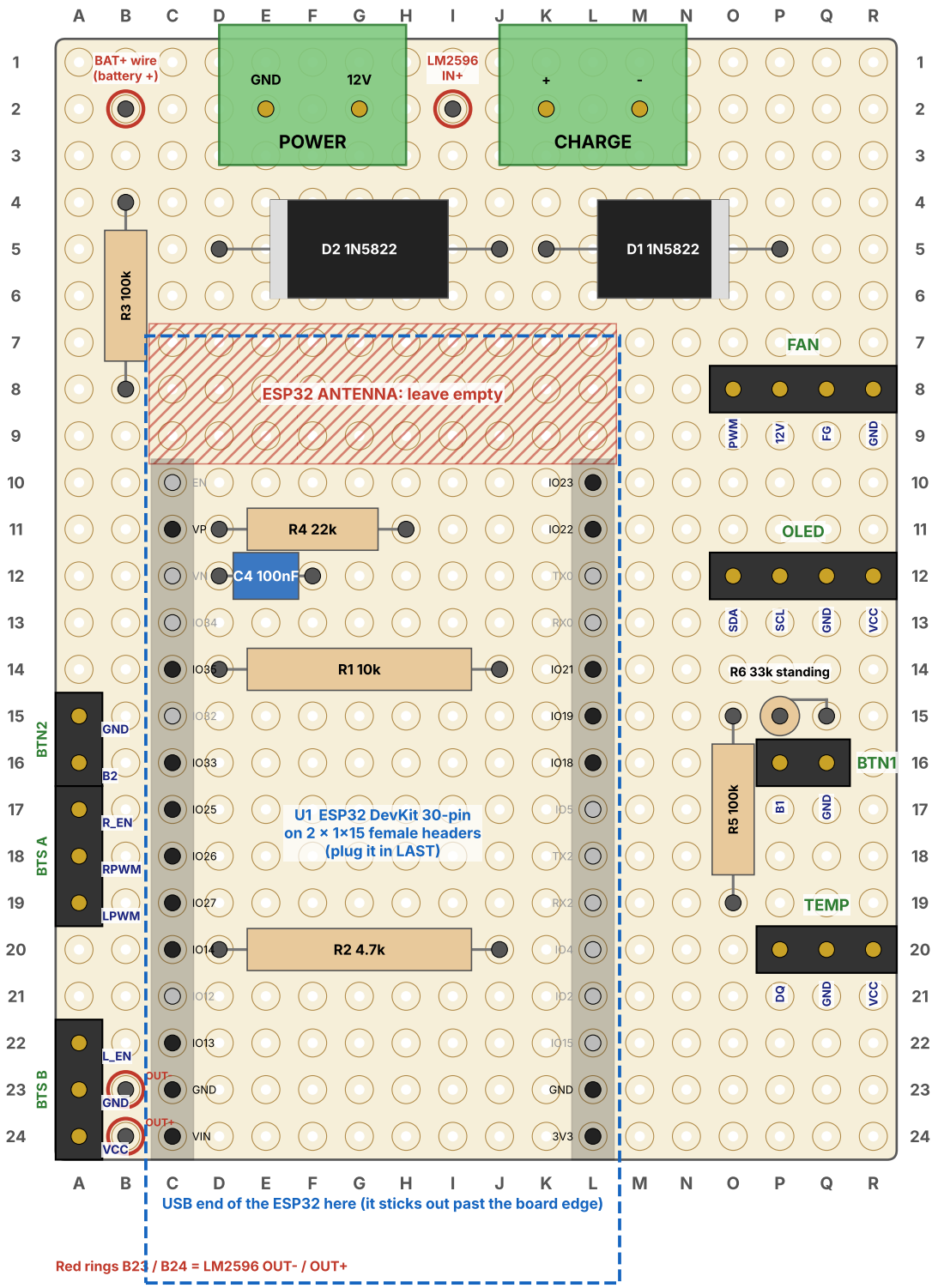
ThermoX UNIVERSAL PCB - 1. PARTS SIDE (top) [ESP32 rows 22.86 mm]

5 × 7 cm perfboard, 18 × 24 holes. Columns A-R, rows 1-24. Mark A, R, 1 and 24 on the board with a marker.



Gold dots = header / terminal pins. Grey dots = part legs. Red rings = wire pads (no part, a wire is soldered here).

R1, R2, R4, C4 lie FLAT under the ESP32 (between the two header rows): solder them before the female headers.

D1, D2 (1N5822): the grey band (stripe) is the cathode. D2 stripe at D5, D1 stripe at P5. Their legs are thick:
open holes D5, J5, K5, P5 with a 1.5 mm drill bit.

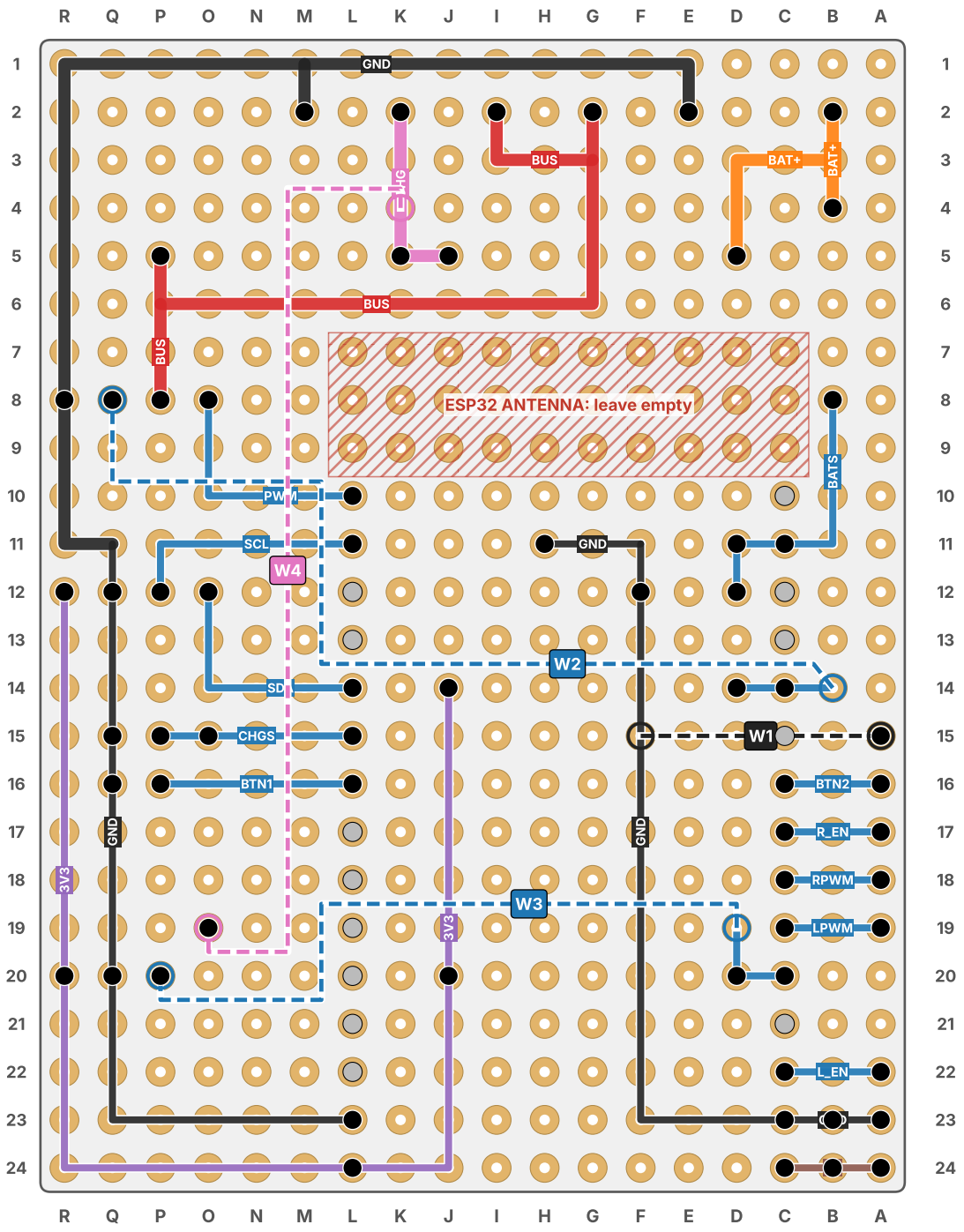
R6 stands up: body over P15, the other leg bent down into Q15.

FAN, OLED, TEMP and BTN1 headers lie ACROSS (left to right). Read the pin names under each pin.

Nothing under the ESP32 antenna (hatched area): no parts and no trails.

ThermoX UNIVERSAL PCB - 2. SOLDER SIDE (bottom) [ESP32 rows 22.86 mm]

Board turned over left to right: column A is now on the RIGHT, row 1 still at the top.



GND: ground
BUS: 12 V after the fuse
BAT+: battery + (gauge, charging)
CHG: charger +
5V: 5 V from the LM2596
3V3: 3.3 V from the ESP32
SIG: signals

Thick lines: bare tinned copper wire 0.8-1 mm (12 V, battery +, charger, top ground). Thin lines: a solder bridge or a bare wire from hole to hole. Black dots are part legs: the trail is soldered to them.

Dashed W1-W4: insulated wire (0.5 mm2 or thinner). They may cross the bare trails. Lay them flat.

Holes next to each other that belong to different colours must NOT touch. Check every joint with a meter.

ThermoX UNIVERSAL PCB - 3. BUILD STEPS [ESP32 rows 22.86 mm]

A. PREPARE

[ ] Use a 5 × 7 cm board (18 × 24 holes). This PDF is for ESP32 pin rows 22.86 mm apart: right header in column L.

[ ] Write A ... R along the top edge and 1 ... 24 down the left edge, on BOTH sides (on the bottom A is on the right).

[ ] Open D5, J5, K5 and P5 with a 1.5 mm drill bit (the 1N5822 legs are thick).

B. PARTS ON TOP (low parts first). Push each leg through, bend it a little, solder, cut.

Step	Part	Holes	Note
[ ] 1	R1 10k	D14 to J14	flat, under the ESP32
[ ] 2	R2 4.7k	D20 to J20	flat, under the ESP32
[ ] 3	R4 22k	D11 to H11	flat, under the ESP32
[ ] 4	C4 100 nF	D12 and F12	under the ESP32, keep it low
[ ] 5	R3 100k	B4 to B8	flat, along column B
[ ] 6	R5 100k	O15 to O19	flat, along column O
[ ] 7	R6 33k	P15 (body) and Q15	standing
[ ] 8	D2 1N5822	J5 to D5	stripe at D5 (open the holes to 1.5 mm)
[ ] 9	D1 1N5822	K5 to P5	stripe at P5 (open the holes to 1.5 mm)
[ ] 10	ESP32 female headers 1x15	C10-C24 and L10-L24	plug the ESP32 in them to align, then solder
[ ] 11	BTN2 2-pin header	A15 (GND), A16 (B2)	
[ ] 12	BTS A 3-pin header	A17 R_EN, A18 RPWM, A19 LPWM	
[ ] 13	BTS B 3-pin header	A22 L_EN, A23 GND, A24 VCC (5 V)	
[ ] 14	FAN 4-pin header	O8 PWM, P8 12V, Q8 FG, R8 GND	lies across
[ ] 15	OLED 4-pin header	O12 SDA, P12 SCL, Q12 GND, R12 VCC	lies across
[ ] 16	BTN1 2-pin header	P16 (B1), Q16 (GND)	lies across
[ ] 17	TEMP 3-pin header	P20 DQ, Q20 GND, R20 VCC (3.3 V)	lies across
[ ] 18	POWER screw terminal 2-pin	E2 GND, G2 12V	wire openings toward the top edge
[ ] 19	CHARGE screw terminal 2-pin	K2 +, M2 -	wire openings toward the top edge

C. TRAILS ON THE BOTTOM (page 2). Follow each line hole by hole; '>' means a corner.

Net	From > corner > ... > to	Wire
[ ] GND	E2 > E1 > R1 > R11 > Q11	bare 0.8-1 mm
[ ] GND	M1 > M2	bare 0.8-1 mm
[ ] GND	A23 > C23	bare / solder
[ ] GND	C23 > F23 > F11	bare / solder
[ ] GND	F11 > H11	bare / solder
[ ] GND	L23 > Q23 > Q11	bare / solder
[ ] BUS	G2 > G3 > I3 > I2	bare 0.8-1 mm
[ ] BUS	G3 > G6 > P6 > P5	bare 0.8-1 mm
[ ] BUS	P6 > P8	bare 0.8-1 mm
[ ] CHG	K2 > K5	bare 0.8-1 mm
[ ] CHG	J5 > K5	bare 0.8-1 mm
[ ] BAT+	B2 > B4	bare 0.8-1 mm
[ ] BAT+	B3 > D3 > D5	bare 0.8-1 mm
[ ] 5V	A24 > C24	bare / solder
[ ] 3V3	L24 > J24 > J14	bare / solder
[ ] 3V3	L24 > R24 > R12	bare / solder
[ ] BATS	B8 > B11 > D11 > D12	bare / solder
[ ] FG	B14 > C14 > D14	bare / solder
[ ] DQ	C20 > D20 > D19	bare / solder
[ ] PWM	L10 > O10 > O8	bare / solder
[ ] SCL	L11 > P11 > P12	bare / solder
[ ] SDA	L14 > O14 > O12	bare / solder
[ ] CHGS	L15 > P15	bare / solder
[ ] BTN1	L16 > P16	bare / solder
[ ] BTN2	A16 > C16	bare / solder
[ ] R_EN	A17 > C17	bare / solder
[ ] RPWM	A18 > C18	bare / solder
[ ] LPWM	A19 > C19	bare / solder
[ ] L_EN	A22 > C22	bare / solder

ThermoX UNIVERSAL PCB - 4. WIRES, CHECKS, FIRST POWER-UP [ESP32 rows 22.86 mm]

D. INSULATED WIRES ON THE BOTTOM (dashed on page 2)

```
[ ] W1  GND  from A15  to F15   (button 2 GND)
[ ] W2  FG   from B14  to Q8    (fan tach)
[ ] W3  DQ   from D19  to P20   (temp sensor data)
[ ] W4  CHG  from K4   to O19   (charger detect)
```

Lay them flat along the path drawn on page 2, not across the ESP32 antenna end.

E. CHECK WITH A METER (beep mode), ESP32 NOT plugged in, nothing connected

```
[ ] Must NOT beep:  B2 (BAT+) to E2 (GND)   G2 (12V) to E2   K2 (CHG) to M2 (GND)   B24 (5V) to B23 (GND)
[ ] Must NOT beep:  R24 (3V3) to Q23 (GND)   B2 to G2   K2 to G2   each pin to its neighbours on the headers
[ ] Must beep:      E2 to M2, E2 to Q12, E2 to R8 (all GND)   G2 to I2, G2 to P8 (12V)   B2 to D5 (BAT+)
[ ] Must beep:      B24 to A24 and C24 (5V)   L24 to R20 and J14 (3V3)   K2 to O19 (CHG, via W4)
```

F. OFF-BOARD WIRING (page 5), then FIRST POWER-UP without the ESP32

[ ] Set the LM2596 to 5.0 V with a meter BEFORE wiring it. Set the USB-C trigger board to 12 V and measure it.

[ ] Wire the LM2596: IN+ to I2, OUT+ to B24, OUT- to B23 (its IN- is the same as OUT-, no wire needed).

[ ] Battery on, switch on BATT: G2 = battery voltage, B24 = 5.0 V, B2 = battery voltage. Switch to OFF: G2 = 0 V.

[ ] Switch OFF, charger in the DC jack: K2 = about 12.6 V, G2 = about 12 V (D1 feeds the board), battery rises.

[ ] Everything OK: switch off, plug in the ESP32 (USB end at the board edge), connect the headers.

G. TEST AND USE

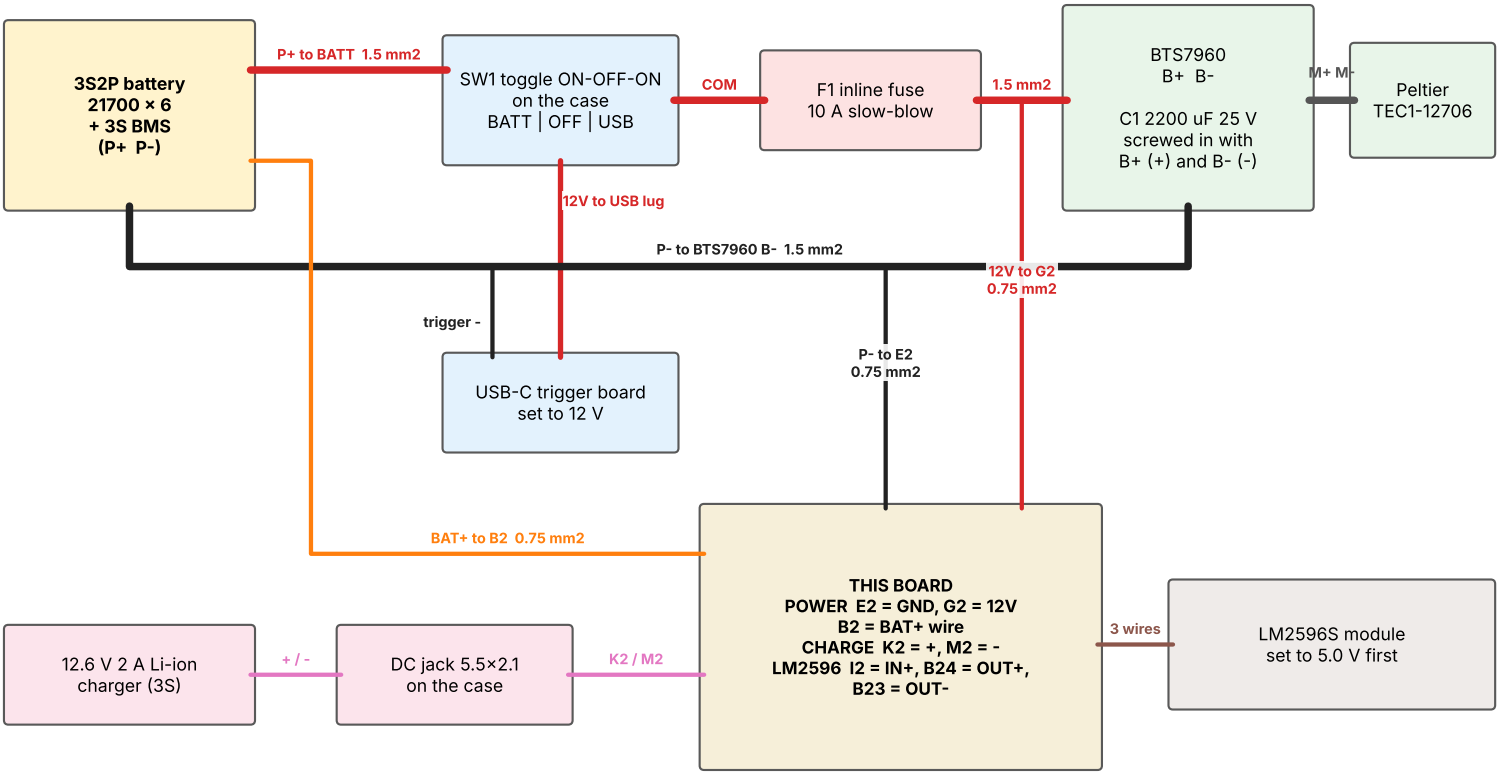
[ ] Upload ThermoX_Bench/ThermoX_Bench.ino, Serial Monitor 115200: check TEMP, BATT, CHARGER, B1/B2, fan (f), Peltier (h, c).

[ ] Upload ThermoX/ThermoX.ino. Set ENABLE_BATTERY_SENSE = true (the battery divider R3/R4 is on this board).
ENABLE_CHARGE_DETECT stays true (R5/R6 are on this board).

[ ] Pins used: Button 1 GPIO18, Button 2 GPIO33, fan PWM GPIO23, fan FG GPIO35, battery GPIO36 (VP),
charger GPIO19, DS18B20 GPIO14, OLED SDA 21 / SCL 22, BTS7960 R_EN 25, RPWM 26, LPWM 27, L_EN 13.

[ ] The DS18B20 runs from 3.3 V on this board (TEMP VCC = 3.3 V). That is fine for the DS18B20.

ThermoX UNIVERSAL PCB - 5. OFF-BOARD WIRING AND PARTS [ESP32 rows 22.86 mm]



Headers to the parts (Dupont wires):

FAN (O8-R8): PWM blue, 12V red, FG yellow, GND black (SUNON 4-wire fan).
OLED (O12-R12): SDA, SCL, GND, VCC (3.3 V). Match the names printed on the OLED, not the order.
TEMP (P20-R20): DQ yellow, GND black, VCC red (DS18B20 probe).
BTN1 (P16-Q16) and BTN2 (A15-A16): one button leg to B1/B2, the other to GND.
BTS A (A17-A19): R_EN, RPWM, LPWM. BTS B (A22-A24): L_EN, GND, VCC (5 V). R_IS and L_IS stay open.

Parts for this board

5 × 7 cm universal PCB (single-sided, 18 × 24 holes) ESP32 DevKit 30-pin + 2 × 1×15 female headers
R1 10k, R2 4.7k, R3 100k, R4 22k, R5 100k, R6 33k (1/4 W) C4 100 nF ceramic D1, D2: 1N5822
2 × 2-pin screw terminal 5.08 mm (KF301-2P) Male headers 2.54 mm: 4+4+3+3+3+2+2 pins
Bare tinned copper wire 0.8-1 mm, thin insulated wire (W1-W4), solder
Off-board: SW1 ON-OFF-ON toggle, inline fuse holder + 10 A slow-blow fuse, C1 2200 uF 25 V, DC jack 5.5×2.1,
LM2596S module, USB-C PD trigger board, 12.6 V 2 A Li-ion charger (3S), wires 1.5 / 0.75 / 0.5 mm2

Safety

The Peltier current (up to about 7 A) goes battery → SW1 → fuse → BTS7960 with thick wire. It does NOT go through this board.
B2 (BAT+) is the battery + before the fuse: keep its wire short and check it never touches ground.
Only one source at a time: never plug the USB-C charger and the DC charger in together.
Charge with SW1 in the middle (OFF). Use a 12.6 V (3S) charger only, never 21 V.
Trigger board at 12 V only. On USB-C keep the Peltier at 40-60 % (phone slider). On battery 80 % or less with the MTS-103.
Not built and tested yet: lay the parts on the board before soldering and check every step with a meter.